

NLTK_I

November 10, 2025

1 NLTK - natural language Processing Kit za Python

- Python alat za manipulaciju prirodnih jezika
- *funkcionalnosti*: pristup korpusima, obradba stringova, označivanje, klasifikacija, komadanje, parsiranje, semantičke interpretacije, ...

```
[7]: !pip install nltk --quiet  
!pip install matplotlib --quiet
```

```
[notice] A new release of pip is available: 25.2 -> 25.3  
[notice] To update, run: C:\Users\Domagoj\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip
```

```
[notice] A new release of pip is available: 25.2 -> 25.3  
[notice] To update, run: C:\Users\Domagoj\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip
```

```
[8]: import nltk
```

Potrebni resursi predstavljaju korpuse tekstova.

```
[2]: nltk.download('gutenberg')  
nltk.download('genesis')  
nltk.download('book')  
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package gutenberg to  
[nltk_data] C:\Users\Domagoj\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\gutenberg.zip.  
[nltk_data] Downloading package genesis to  
[nltk_data] C:\Users\Domagoj\AppData\Roaming\nltk_data...  
[nltk_data] Unzipping corpora\genesis.zip.  
[nltk_data] Downloading collection 'book'  
[nltk_data] |  
[nltk_data] | Downloading package abc to  
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...  
[nltk_data] | Unzipping corpora\abc.zip.
```

```

[nltk_data] | Downloading package brown to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\brown.zip.
[nltk_data] | Downloading package chat80 to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\chat80.zip.
[nltk_data] | Downloading package cmudict to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\cmudict.zip.
[nltk_data] | Downloading package conll2000 to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\conll2000.zip.
[nltk_data] | Downloading package conll2002 to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\conll2002.zip.
[nltk_data] | Downloading package dependency_treebank to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\dependency_treebank.zip.
[nltk_data] | Downloading package genesis to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Package genesis is already up-to-date!
[nltk_data] | Downloading package gutenber to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Package gutenber is already up-to-date!
[nltk_data] | Downloading package ieer to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\ieer.zip.
[nltk_data] | Downloading package inaugural to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Package inaugural is already up-to-date!
[nltk_data] | Downloading package movie_reviews to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\movie_reviews.zip.
[nltk_data] | Downloading package nps_chat to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\nps_chat.zip.
[nltk_data] | Downloading package names to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Package names is already up-to-date!
[nltk_data] | Downloading package ppattach to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\ppattach.zip.
[nltk_data] | Downloading package reuters to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Downloading package senseval to
[nltk_data] |   C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\senseval.zip.
[nltk_data] | Downloading package state_union to

```

```

[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\state_union.zip.
[nltk_data] | Downloading package stopwords to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\stopwords.zip.
[nltk_data] | Downloading package swadesh to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\swadesh.zip.
[nltk_data] | Downloading package timit to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\timit.zip.
[nltk_data] | Downloading package treebank to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\treebank.zip.
[nltk_data] | Downloading package toolbox to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\toolbox.zip.
[nltk_data] | Downloading package udhr to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\udhr.zip.
[nltk_data] | Downloading package udhr2 to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\udhr2.zip.
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\unicode_samples.zip.
[nltk_data] | Downloading package webtext to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\webtext.zip.
[nltk_data] | Downloading package wordnet to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Downloading package wordnet_ic to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\wordnet_ic.zip.
[nltk_data] | Downloading package words to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\words.zip.
[nltk_data] | Downloading package maxent_treebank_pos_tagger to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping taggers\maxent_treebank_pos_tagger.zip.
[nltk_data] | Downloading package maxent_ne_chunker to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping chunkers\maxent_ne_chunker.zip.
[nltk_data] | Downloading package universal_tagset to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping taggers\universal_tagset.zip.
[nltk_data] | Downloading package punkt to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...

```

```

[nltk_data] | Package punkt is already up-to-date!
[nltk_data] | Downloading package book_grammars to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping grammars\book_grammars.zip.
[nltk_data] | Downloading package city_database to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping corpora\city_database.zip.
[nltk_data] | Downloading package tagsets to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping help\tagsets.zip.
[nltk_data] | Downloading package panlex_swadesh to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Downloading package averaged_perceptron_tagger to
[nltk_data] | C:\Users\Domagoj\AppData\Roaming\nltk_data...
[nltk_data] | Unzipping taggers\averaged_perceptron_tagger.zip.
[nltk_data] |
[nltk_data] Done downloading collection book
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\Domagoj\AppData\Roaming\nltk_data...

```

[2]: True

```

[9]: # dohvatiti primjere koji se nalaze u knjizi: Bird et al.,NLTK with Python
      #!python3 -m nltk.downloader book

```

```

from nltk import FreqDist
from nltk.book import text1,text2,text3

```

```

*** Introductory Examples for the NLTK Book ***
Loading text1, ..., text9 and sent1, ..., sent9
Type the name of the text or sentence to view it.
Type: 'texts()' or 'sents()' to list the materials.
text1: Moby Dick by Herman Melville 1851
text2: Sense and Sensibility by Jane Austen 1811
text3: The Book of Genesis
text4: Inaugural Address Corpus
text5: Chat Corpus
text6: Monty Python and the Holy Grail
text7: Wall Street Journal
text8: Personals Corpus
text9: The Man Who Was Thursday by G . K . Chesterton 1908

```

```

[10]: # konkordancija
      print(text1)
      print('*** konkordacija:')
      text1.concordance('monstrous') # izlistaj kontekste s riječju 'monstrous'
      print('*** konkordacija sa sličnim pojmovima:')
      text1.similar('monstrous') # pronadi slične kontekste riječju 'monstrous'

```

```

print(text2)
print("*** zajednički konteksti:")
text2.common_contexts(["monstrous", "very"]) # dijeljeni konteksti?

print(text3)
print("*** položaj riječi u tekstu:")
text3.dispersion_plot(["God", "Adam", "LORD", "sin", "apple"]) # kada se koja_
↪ riječ pojavila
fdist3 = FreqDist(text3) # izračunati frekvenciju riječi u tekstu
fdist3.plot(10) # prikazati prvih 10 pojmova

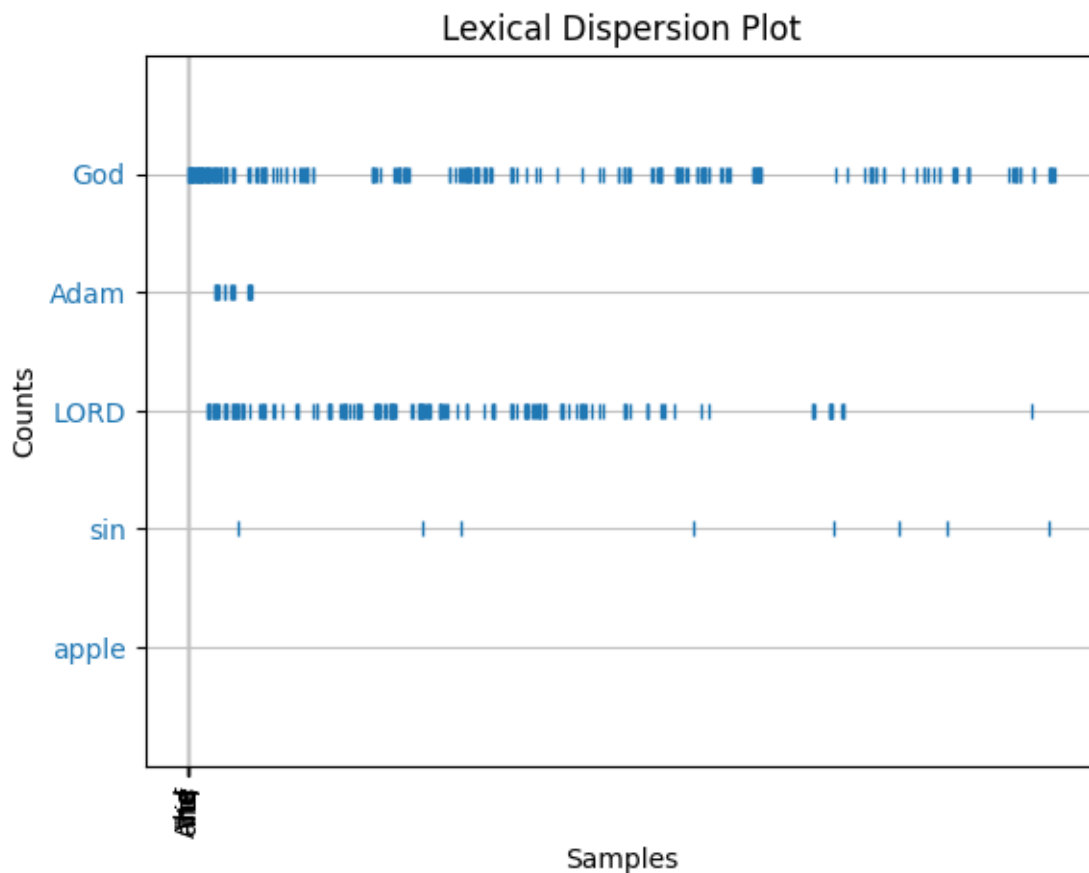
```

```

<Text: Moby Dick by Herman Melville 1851>
*** konkordacija:
Displaying 11 of 11 matches:
ong the former , one was of a most monstrous size . ... This came towards us ,
ON OF THE PSALMS . " Touching that monstrous bulk of the whale or ork we have r
ll over with a heathenish array of monstrous clubs and spears . Some were thick
d as you gazed , and wondered what monstrous cannibal and savage could ever hav
that has survived the flood ; most monstrous and most mountainous ! That Himmal
they might scout at Moby Dick as a monstrous fable , or still worse and more de
th of Radney .' " CHAPTER 55 Of the Monstrous Pictures of Whales . I shall ere l
ing Scenes . In connexion with the monstrous pictures of whales , I am strongly
ere to enter upon those still more monstrous stories of them which are to be fo
ght have been rummaged out of this monstrous cabinet there is no telling . But
of Whale - Bones ; for Whales of a monstrous size are oftentimes cast up dead u
*** konkordacija sa sličnim pojmovima:
true contemptible christian abundant few part mean careful puzzled
mystifying passing curious loving wise doleful gamesome singular
delightfully perilous fearless
<Text: Sense and Sensibility by Jane Austen 1811>
*** zajednički konteksti:
am_glad a_pretty a_lucky is_pretty be_glad
<Text: The Book of Genesis>
*** položaj riječi u tekstu:

```

[10]: <Axes: title={'center': 'Lexical Dispersion Plot'}, xlabel='Samples', ylabel='Counts'>



2 NLTK korpusi

Korpus predstavlja skup tekstova prirodnog govora popraćen lingvističkim metapodacima: * NLTK obuhvaća različit skup korpusa koji se mogu učitati koristeći `nltk.corpus` paket. Svakom korpusu se pristupa preko `corpus reader` objekta iz `nltk.corpus`. * Korpusi se sastoje od skupa datoteka (dokumenata) i njihovih identifikatora: * `inaugural` - govori američkih predsjednika * `Brown` - prvi korpus Brown sveučilišta od milijun riječi, 500 izvora (news, editorial, hobi, ...), označen i kategoriziran * `Penn Treebank` korpus označen i parsiran * dohvaćanje podataka može biti preko riječi, rečenica, odlomaka, kategorija, ...

```
[11]: # govor predsjednika SAD-a
from nltk.corpus import inaugural
# popis datoteka
inaugural.fileids()

# jedan od tekstova
WashSpeech = '1789-Washington.txt'
inaugural.raw(WashSpeech)
# tokenizacija po rijecima
```

```

inaugural.words(WashSpeech) # lista
# tokenizacija po recenicima (+rijeci)
inaugural.sents(WashSpeech) # lista listi
# tokenizacija po odjeljcima (+rijeci)
inaugural.paras(WashSpeech)

```

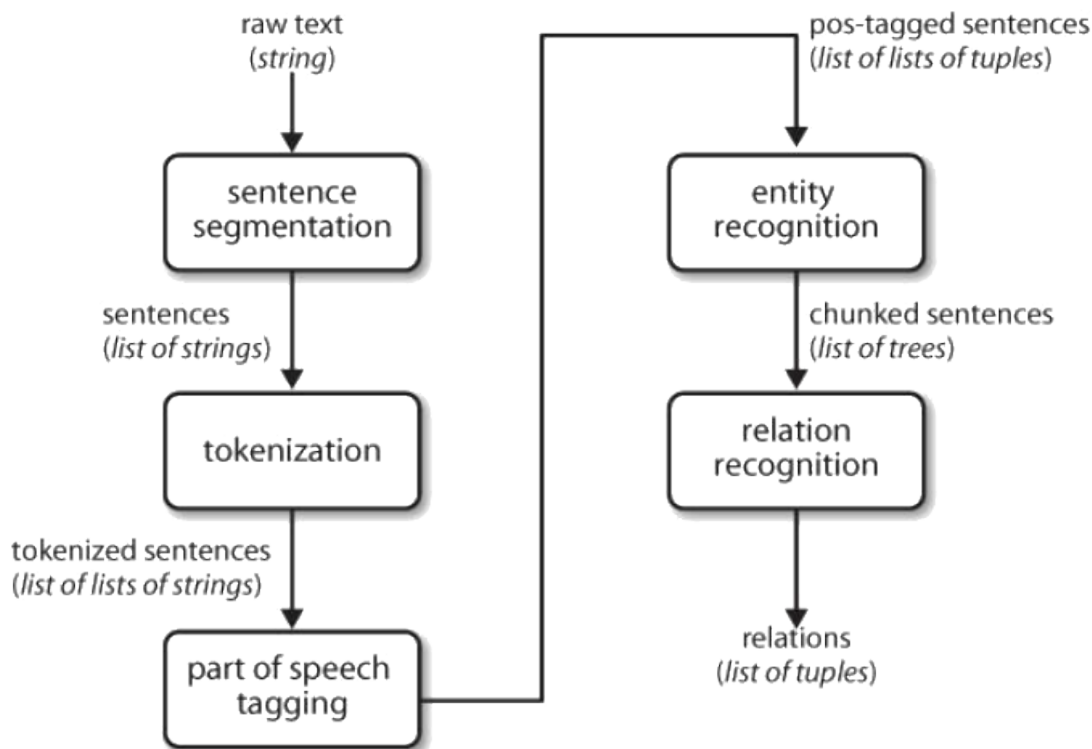
```

[11]: [[['Fellow', '-', 'Citizens', 'of', 'the', 'Senate', 'and', 'of', 'the',
'House', 'of', 'Representatives', ':'], ['Among', 'the', 'vicissitudes',
'incident', 'to', 'life', 'no', 'event', 'could', 'have', 'filled', 'me',
'with', 'greater', 'anxieties', 'than', 'that', 'of', 'which', 'the',
'notification', 'was', 'transmitted', 'by', 'your', 'order', ',', 'and',
'received', 'on', 'the', '14th', 'day', 'of', 'the', 'present', 'month', '.'],
['On', 'the', 'one', 'hand', ',', 'I', 'was', 'summoned', 'by', 'my', 'Country',
',', 'whose', 'voice', 'I', 'can', 'never', 'hear', 'but', 'with', 'veneration',
'and', 'love', ',', 'from', 'a', 'retreat', 'which', 'I', 'had', 'chosen',
'with', 'the', 'fondest', 'predilection', ',', 'and', ',', 'in', 'my',
'flattering', 'hopes', ',', 'with', 'an', 'immutable', 'decision', ',', 'as',
'the', 'asylum', 'of', 'my', 'declining', 'years', '--', 'a', 'retreat',
'which', 'was', 'rendered', 'every', 'day', 'more', 'necessary', 'as', 'well',
'as', 'more', 'dear', 'to', 'me', 'by', 'the', 'addition', 'of', 'habit', 'to',
'inclination', ',', 'and', 'of', 'frequent', 'interruptions', 'in', 'my',
'health', 'to', 'the', 'gradual', 'waste', 'committed', 'on', 'it', 'by',
'time', '.'], ['On', 'the', 'other', 'hand', ',', 'the', 'magnitude', 'and',
'difficulty', 'of', 'the', 'trust', 'to', 'which', 'the', 'voice', 'of', 'my',
'country', 'called', 'me', ',', 'being', 'sufficient', 'to', 'awaken', 'in',
'the', 'wisest', 'and', 'most', 'experienced', 'of', 'her', 'citizens', 'a',
'distrustful', 'scrutiny', 'into', 'his', 'qualifications', ',', 'could', 'not',
'but', 'overwhelm', 'with', 'despondence', 'one', 'who', '(', 'inheriting',
'inferior', 'endowments', 'from', 'nature', 'and', 'unpracticed', 'in', 'the',
'duties', 'of', 'civil', 'administration', ')', 'ought', 'to', 'be',
'peculiarly', 'conscious', 'of', 'his', 'own', 'deficiencies', '.'], ['In',
'this', 'conflict', 'of', 'emotions', 'all', 'I', 'dare', 'aver', 'is', 'that',
'it', 'has', 'been', 'my', 'faithful', 'study', 'to', 'collect', 'my', 'duty',
'from', 'a', 'just', 'appreciation', 'of', 'every', 'circumstance', 'by',
'which', 'it', 'might', 'be', 'affected', '.'], ['All', 'I', 'dare', 'hope',
'is', 'that', 'if', ',', 'in', 'executing', 'this', 'task', ',', 'I', 'have',
'been', 'too', 'much', 'swayed', 'by', 'a', 'grateful', 'remembrance', 'of',
'former', 'instances', ',', 'or', 'by', 'an', 'affectionate', 'sensitivity',
'to', 'this', 'transcendent', 'proof', 'of', 'the', 'confidence', 'of', 'my',
'fellow', 'citizens', ',', 'and', 'have', 'thence', 'too', 'little',
'consulted', 'my', 'incapacity', 'as', 'well', 'as', 'disinclination', 'for',
'the', 'weighty', 'and', 'untried', 'cares', 'before', 'me', ',', 'my', 'error',
'will', 'be', 'palliated', 'by', 'the', 'motives', 'which', 'mislead', 'me',
',', 'and', 'its', 'consequences', 'be', 'judged', 'by', 'my', 'country',
'with', 'some', 'share', 'of', 'the', 'partiality', 'in', 'which', 'they',
'originated', '.']], ...]

```

3 Obrada teksta

Danas obrađujemo: 1. raščlamba odlomaka u rečenice 2. proces dijeljenja niza riječi i znakova interpunkcije u listu dijelova – engl. *tokenization* 3. Pridruživanje oznake riječima – engl. *tagging*



3.1 Raščlambe (engl. tokenization)

- različite vrste tokenizacije koje mogu ovisiti o jeziku
- rečenične, riječne i regex raščlambe
- funkcionalnosti dostupne u paketu `nltk.tokenize`

```
[7]: # neki specifični dijelovi NLTK se moraju preuzeti naknadno ...
# python3 -m nltk.downloader punkt
import nltk
# prije toga u terminalu povuci korpus za tokenizaciju imenom 'punkt'
# nltk.download() # upisati 'punkt'
sentenceEN = "Hello World. It's good to see you. Thanks for buying this book."
sentenceSP = "Hola amigo. Estoy bien."

# rečenična tokenizacija
from nltk.tokenize import sent_tokenize
sent_tokenize(sentenceEN);
sent_tokenize(sentenceSP);

# što je ekvivalentno s ...
```



```

# tokenizacija za ENG jezik
english_tokenizer = nltk.data.load('tokenizers/punkt/english.pickle')
english_tokenizer.tokenize(sentenceEN);

# tokenizacija za SP jezik
spanish_tokenizer = nltk.data.load('tokenizers/punkt/spanish.pickle')
spanish_tokenizer.tokenize(sentenceSP);

# tokenizacija po riječima
from nltk.tokenize import word_tokenize
word_tokenize(sentenceEN);

# mogu li se koristiti različiti tokenizatori?
# WordPunctTokenizer
from nltk.tokenize import WordPunctTokenizer
tokenizer = WordPunctTokenizer()
tokenizer.tokenize("Can't is a contraction.")

# TreeBank (NLTK-ov standardni tokenizator)
from nltk.tokenize import TreebankWordTokenizer
tokenizer = TreebankWordTokenizer()
tokenizer.tokenize("Can't is a contraction.")

# regex tokenizator
from nltk.tokenize import regexp_tokenize
regexp_tokenize("Can't is a contraction.", r"[\w']+")

```

```
[7]: ["Can't", 'is', 'a', 'contraction']
```

3.2 Korjenovanje (engl. stemming)

Tehnika odstranjivanja afiksa iz riječi, tako da ostane samo korijen. Nekoliko *stemming* algoritama podržano u NLTK paketu `nltk.stem`: * *Porter Stemmer algoritam* - uklanja sufikse i određuje korjene engleskih riječi * *Lancaster stemming algoritam* - Lancaster University korjenovatelj algoritam * *Snowball stemmer* - korjenovatelj s primjenom na druge jezike * vlastito definirani regex temeljeni korjenovatelj

```

[8]: # Porter stemmer
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
stemmer.stem('cooking')
stemmer.stem('cookery')
# Lancaster stemmer
from nltk.stem import LancasterStemmer
stemmer = LancasterStemmer()

```

```

stemmer.stem('cooking')
stemmer.stem('cookery')
# regex stemmer
from nltk.stem import RegexpStemmer
stemmer = RegexpStemmer('ing') # zamjeniti s 'ing$'
stemmer.stem('cooking')
stemmer.stem('cookery')
stemmer.stem('ingleside') # kako ovo ispraviti?
from nltk.stem import SnowballStemmer
# podrzani jezici u SnowballStemmer
SnowballStemmer.languages

```

```

[8]: ('arabic',
      'danish',
      'dutch',
      'english',
      'finnish',
      'french',
      'german',
      'hungarian',
      'italian',
      'norwegian',
      'porter',
      'portuguese',
      'romanian',
      'russian',
      'spanish',
      'swedish')

```

3.3 Lematizacija (engl. lemmatization)

- *lema* ili *leksem* = kanonski oblik riječi
- *morf* = korijen riječi
- Postupak sličan korjenovanju, teži utvrđivanju leme riječi

```

[9]: from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
lemmatizer.lemmatize('cooking') # glagolska imenica
lemmatizer.lemmatize('cooking', pos='v') # naglasavamo da je rijec 'verb'=glagol
lemmatizer.lemmatize('cookbooks') # lema od množine 'kuharice'

# usporedimo korjenovatelj s lematizatorom
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
stemmer.stem('believes')
lemmatizer.lemmatize('believes')
stemmer.stem('buses')

```

```

lemmatizer.lemmatize('buses')
stemmer.stem('bus')
lemmatizer.lemmatize('bus')

```

[9]: 'bus'

4 N-gramski jezični model

N-gramski jezični modeli su statistički modeli koji procjenjuju vjerojatnost pojave riječi na temelju prethodnih (n-1) riječi u tekstu. Najčešće korišteni modeli su bigrami (n=2) i trigrami (n=3). Ovi modeli pretpostavljaju da je vjerojatnost pojave riječi uvjetovana samo ograničenim brojem prethodnih riječi, čime se pojednostavljuje modeliranje jezika i omogućuje automatska obrada prirodnog jezika, poput predviđanja riječi, automatskog prevodenja i prepoznavanja govora.

```

[11]: import nltk
      from nltk.corpus import PlaintextCorpusReader
      from nltk.tokenize import sent_tokenize
      from nltk.lm.preprocessing import pad_both_ends, flatten,
      ↪ padded_everygram_pipeline
      from nltk.util import bigrams, ngrams
      from nltk.lm import MLE, Laplace, Vocabulary
      import nltk.lm as lm
      import random
      import re

```

```

[20]: # trivijalni tekst podjeljen u tokene
      text = [['a', 'b', 'c'], ['a', 'c', 'd', 'c', 'e', 'f'], ['a', 'f', 'e']]
      test = [('a', 'b'), ('c', 'd')]

```

```

[22]: # izgradnja ngrama
      from nltk.util import bigrams, ngrams
      list(bigrams(text[0]))
      [list(ngrams(sent, n=3)) for sent in text]

```

```

[22]: [['a', 'b', 'c'],
      [('a', 'c', 'd'), ('c', 'd', 'c'), ('d', 'c', 'e'), ('c', 'e', 'f')],
      [['a', 'f', 'e']]

```

```

[23]: # dodavanje nadopuna (engl. paddings)
      from nltk.util import pad_sequence
      list(pad_sequence(text[0],
      pad_left=True,
      left_pad_symbol="<s>",
      pad_right=True,
      right_pad_symbol="</s>",
      n=2))

```

```
[23]: ['<s>', 'a', 'b', 'c', '</s>']
```

```
[24]: list(bigrams(pad_both_ends(text[0], n=2)))
```

```
[24]: [('<s>', 'a'), ('a', 'b'), ('b', 'c'), ('c', '</s>')]
```

Za model nam je potreban skup svih tokena kao vokabular.

```
[25]: # koristimo gotov alat iz nltk.lm za dodavanje nadopuna
from nltk.lm.preprocessing import flatten
vocab = list(flatten(pad_both_ends(sent, n=2) for sent in text))
vocab
```

```
[25]: ['<s>',
'a',
'b',
'c',
'</s>',
'<s>',
'a',
'c',
'd',
'c',
'e',
'f',
'</s>',
'<s>',
'a',
'f',
'e',
'</s>']
```

```
[ ]: from nltk.lm.preprocessing import padded_everygram_pipeline

train, vocab = padded_everygram_pipeline(2, text)
#train = [list(ngrams(pad_both_ends(sent, n=2), 2)) for sent in text]
#vocab = list(flatten(pad_both_ends(sent, n=2) for sent in text))

print('text', text)
# koristimo pipeline u kojem su train i vocab generatori
print('train', list(list(train)[0]))
print('vocab', list(vocab))
```

```
text [['a', 'b', 'c'], ['a', 'c', 'd', 'c', 'e', 'f'], ['a', 'f', 'e']]
train [('<s>',), ('<s>', 'a'), ('a',), ('a', 'b'), ('b',), ('b', 'c'), ('c',),
('c', '</s>'), ('</s>',)]
vocab ['<s>', 'a', 'b', 'c', '</s>', '<s>', 'a', 'c', 'd', 'c', 'e', 'f',
'</s>', '<s>', 'a', 'f', 'e', '</s>']
```

```
[28]: text = [['a', 'b', 'c'], ['a', 'c', 'd', 'c', 'e', 'f'], ['a', 'f', 'e']]
      test = [('a', 'b'), ('c', 'd')]

      from nltk.lm.preprocessing import padded_everygram_pipeline
      train, vocab = padded_everygram_pipeline(2, text)

      # standardni MLE bigram model
      from nltk.lm import MLE
      lm = MLE(2)

      # treniranje modela
      lm.fit(train, vocab)

      # koliko puta se pojavilo 'a'
      lm.counts['a']

      # koliko se puta pojavilo 'a b' kao bigram
      lm.counts[['a']]['b']

      # vjerojatnost pridružena modelom
      lm.score('a')

      # koliko puta se pojavio 'b' nakon 'a' kao bigram
      lm.score("b", ["a"])

      lm.entropy(test)
      lm.perplexity(test)
```

[28]: 3.0000000000000004

4.1 Primjer s korpusom teksta

```
[34]: def prepare(corpora, split=0.7):
      dataset = open(corpora, 'r', encoding='utf-8').read()
      #dataset = re.sub('\n', dataset, '.')
      sents = nltk.tokenize.sent_tokenize(dataset)
      #pdb.set_trace()
      random.shuffle(sents)
      index = random.randint(0, int(split*len(sents)))
      trainset, testset = sents[index:], sents[:index]

      # store training set
      with open('data/train.txt', 'w', encoding='utf-8') as train:
          for sent in trainset:
              train.write(sent)
              train.write('\n')
```

```
# store test set
with open('data/test.txt','w', encoding='utf-8') as test:
    for sent in testset:
        test.write(sent)
        test.write('\n')
```

```
[ ]: prepare('data/werther.txt')
```

```
[36]: PlaintextCorpusReader('data/', 'train.txt').sents()
```

```
[36]: [['Zar', 'si', 'ti', ',', 'koji', 'si', 'dosta', 'siromašnim', 'stvorio',
'čovjeka', ',', 'morao', 'njemu', 'dati', 'i', 'braću', ',', 'koja', 'mu',
'otimlju', 'još', 'i', 'onu', 'mrvicu', 'siromaštva', ',', 'mrvicu',
'pouzdanja', 'koje', 'ima', 'u', 'tebe', ',', 'u', 'tebe', ',', 'Sveljubeći',
'!'], ['Ujutro', ',', 'oko', 'šeste', ',', 'uđe', 'u', 'sobu', 'sluga', 'sa',
'svijećom', '.'], ...]
```

```
[37]: PlaintextCorpusReader('data/', 'test.txt').sents()
```

```
[37]: [['Kako', 'je', 'ukorio', 'moju', 'previše', 'veliku', 'osjetljivost', ',',
'kako', 'moje', 'prenapete', 'ideje', 'o', 'djelovanju', ',', 'utjecaju', 'na',
'ostale', ',', 'pronicanju', 'posala', ',', 'donekle', 'štuje', 'kano',
'mladenačku', 'dobru', 'volju', ',', 'pa', 'ne', 'misli', 'da', 'ih',
'iskorijeni', ',', 'gdje', 'je', 'za', 'njih', 'pravo', 'mjesto', 'i', 'gdje',
'mogu', 'djelovati', 'najsnažnije', '.'], ['Goethe', 'se', 'u', 'Weimaru',
'bavio', 'administrativnim', 'poslovima', ',', 'upravljao', 'je', 'financijama',
'i', 'javnim', 'službama', ',', 'te', 'je', 'od', '1791', '.', 'vodio', 'i',
'dvorsko', 'kazalište', '.'], ...]
```

```
[39]: train_set = PlaintextCorpusReader('data/', 'train.txt').sents()
test_set = PlaintextCorpusReader('data/', 'test.txt').sents()
```

```
# ngrams
nn = 2

# normaliziraj tokene
train_set = [[token.lower() for token in sent] for sent in train_set ]
test_set = [[token.lower() for token in sent] for sent in test_set ]

test = [list(ngrams(sent,nn)) for sent in test_set][0]

from nltk.lm.preprocessing import padded_everygram_pipeline
train, vocab = padded_everygram_pipeline(nn, train_set)
```

```
[40]: train_set[0]
test_set[0]
```

[40]: ['kako',
 'je',
 'ukorio',
 'moju',
 'previše',
 'veliku',
 'osjetljivost',
 ',',
 'kako',
 'moje',
 'prenapete',
 'ideje',
 'o',
 'djelovanju',
 ',',
 'utjecaju',
 'na',
 'ostale',
 ',',
 'pronicanju',
 'posala',
 ',',
 'donekle',
 'štuje',
 'kano',
 'mladenačku',
 'dobru',
 'volju',
 ',',
 'pa',
 'ne',
 'misli',
 'da',
 'ih',
 'iskorijeni',
 ',',
 'gdje',
 'je',
 'za',
 'njih',
 'pravo',
 'mjesto',
 'i',
 'gdje',
 'mogu',
 'djelovati',
 'najsnažnije',

```
['.']
```

Treniranje primjera iz vlastitog korpusa.

```
[41]: from nltk.lm import MLE, Laplace
lm = MLE(nn)

# treniraj model
lm.fit(train, vocab)

lm.vocab.lookup('bio'),
lm.counts[['bio']]['je']

lm.entropy(test)
lm.perplexity(test)
lm.generate(10)
```

```
[41]: ['</s>',
      'uvijek',
      'imali',
      'i',
      'kolo',
      'sreće',
      'nikakvom',
      'hirovitom',
      'neučtivošću',
      ';']
```

```
[ ]:
```